Haofei Liang

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Education

Sept. 2022 – Dec. 2026 (expected) **Shanghai Jiao Tong University**
Doctor of Engineering candidate in Computer Science and Technology.
Research area: applications and optimization of fully homomorphic encryption.
Advisor: Prof. Yu Yu

Sept. 2020 – Aug. 2022 **Shanghai Jiao Tong University**
Master's study in Computer Science and Technology; transferred to the Doctor of Engineering program in 2022.
Advisor: Prof. Yu Yu

Sept. 2015 – June 2019 **Wuhan University**
B.Sc. in Information Security.

Research Interests

My research focuses on **fully homomorphic encryption (FHE)**, especially the **TFHE** family of schemes. Specific topics include:

Functional and Programmable Bootstrapping Evaluating functions during bootstrapping while reducing latency.

Circuit Bootstrapping Efficient conversion between FHE ciphertext formats for deeper homomorphic computation.

Applications of FHE Privacy-preserving protocols including oblivious message retrieval and single secret leader election.

Publications

Conference Proceedings

- **Haofei Liang**, Zeyu Liu, Eran Tromer, Xiang Xie, and Yu Yu. *InstantOMR: Oblivious Message Retrieval with Low Latency and Optimal Parallelizability*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. (Published) USENIX · ePrint: 2025/2317.
- **Haofei Liang**, Zeyu Liu, Yunhao Wang, Xiang Xie, Yu Yu, and Fan Zhang. *Relect: Single Secret Leader Election via FHE with Reduced Computation and Communication and Transparent Setup*. In: 33rd ACM Conference on Computer and Communications Security (ACM CCS 2026), The Hague, The Netherlands, November 15–19, 2026. (Accepted, to appear) ACM CCS · ePrint: 2026/1619.
- Zhenkai Hu, **Haofei Liang**, Xiao Wang, Xiang Xie, Kang Yang, Yu Yu, and Wenhao Zhang. *Ajax: Fast Threshold Fully Homomorphic Encryption without Noise Flooding*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. (Published) USENIX · ePrint: 2025/1834.
- Fengrun Liu, **Haofei Liang**, Xiang Xie, Yu Yu, Wenting Zheng, and Yuncong Hu. *HasteBoots: Proving TFHE Programmable Bootstrapping in Seconds*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. (Published) USENIX · ePrint: 2025/261.

Conference Presentations

Aug. 2026 **USENIX Security 2026**, Baltimore, MD, USA – presented *InstantOMR*.
Nov. 2026 **ACM CCS 2026**, The Hague, The Netherlands – scheduled presentation of *Relect*.

Open-Source Projects

- **primus-fhe** – Primary open-source Rust library for high-performance TFHE research and implementations, including NTT-based primitives; lead developer and maintainer. Reusable implementations developed in *primus* are being consolidated into this library.
- **InstantOMR** – Reference implementation of low-latency, optimally parallelizable oblivious message retrieval; lead developer and first author of the corresponding paper.
- **Ajax** – Reference implementation of threshold fully homomorphic encryption without noise flooding; primary developer.
- **HasteBoots** – High-performance zero-knowledge proofs for TFHE programmable bootstrapping; primary developer.
- **Relect** – Single secret leader election with transparent setup and reduced computation and communication; lead developer and first author of the corresponding paper.

Skills

Programming and Engineering Rust for FHE implementation since 2022; Python for data analysis, benchmarking, and figure generation; C for supporting implementations; performance profiling and parallel optimization.
Cryptography TFHE functional, programmable, and circuit bootstrapping; LWE/RLWE techniques; NTT-based polynomial arithmetic; noise analysis and parameter selection; working knowledge of BFV, BGV, and CKKS.
Research Tools Git, Linux, Typst, and LaTeX; reproducible benchmarking and academic writing workflows.
Languages Native Mandarin Chinese; English (CET-6; proficient reading and writing, conversational speaking).

Selected Links

- **Website:** haofeiliang.github.io
- **Publications:** haofeiliang.github.io/publications/
- **Projects:** haofeiliang.github.io/projects/
- **GitHub:** github.com/haofeiliang